Esp32Tap — Depth-First Cluster Build and Test

Guide mode: Empty-board build · Build one cluster, prove it, sign it, then unlock the next.

No treadmill cable before clusters 1–10 pass; do not connect a treadmill cable until Cluster 11. No exact physical placement is prescribed. Follow the named parts, pins, nets, and colors.

Device serial:	**Operator:**
Hardware revision:	**Date:**

Source manifest: esp32tap-breadboard-netlist.csv; esp32tap-breadboard-clusters.html; esp32tap-breadboard-from-to.html; DigiKey invoice 129832848; superseded module checklist used only for already-proven limits and sequences.

Source contract

Mutually exclusive source rule: USB and STANDALONE POWER are mutually exclusive—one source only. USB connected means STANDALONE POWER physically removed. STANDALONE POWER installed means USB physically disconnected. All sources off before any RJ45 attachment/removal, harness change, or direct-path change. COIL POWER stays open through the unloaded test.

Safety contract

- Initial source: 8.00 V, initial 250 mA current limit; coil-open current below 50 mA. VIN must be 7.20–7.90 V and LOGIC_3V3 must be 3.20–3.40 V.
- TREAD_OK is low below UV, high at 8.00 V, and low above OV. UV rising is 6.25–6.55 V; OV falling is 10.30–10.90 V.
- TPS709 enabled: 4.75–5.25 V; TPS709 disabled: below 0.25 V. Loaded relay current limit no more than 500 mA.
- Coil: 90–110 mA and at least 4.50 V. BC337 VCE no more than 0.30 V. Feedback (1,0) energized and (0,1) bypass; 00 or 11 is a fault.
- Removal of USB logic power, GPIO21 command, TREAD_OK, or VIN must restore NC bypass within 100 ms.
- Five-minute hold: TPS709 and BC337 no more than 45°C and no more than 10°C over ambient.
- Any unexpected reset is a STOP. No treadmill relay transfer or MCU TX is permitted by this guide.

Firmware: The current no-control diagnostic cannot pass a relay exercise gate and cannot pass a UART exercise gate. Relay/coil/TX work requires a bounded relay exerciser with exact identity. Standalone/bypass requires an observer identity and manifest proving relay and TX disabled.

Wire palette: BLACK GND · RED +8V_RAW/FUSED_8V · ORANGE VIN · BLUE TSR_3V3/LOGIC_3V3 · VIOLET +5V_RLY/coil · YELLOW control/UART · GREEN feedback/RX · WHITE direct pass-through.

Cluster 1 — Raw protection

Dependencies: Empty board and verified bench equipment **Inputs:** +8V_RAW, GND	**Outputs:** VIN, FUSED_8V **Source state:** USB absent; STANDALONE POWER removed; COIL POWER removed; bench 8.00 V, initial 250 mA limit

Parts

F1 RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar. **D1** 1N5822-TP Schottky · 1N5822-TPMSCT-ND · cathode stripe faces VIN. **D2** P6KE12A-TP · P6KE12A-TPMSCT-ND · cathode to VIN. **CIN** 22 µF 25 V · 732-8821-1-ND · negative stripe to GND.

Operator part record: F1 · RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar
Operator wiring record: F1 · lead 1 · +8V_RAW · RED; F1 · lead 2 · FUSED_8V · RED; D1 · anode · FUSED_8V · RED; D1 · cathode · VIN · ORANGE; D2 · cathode · VIN · ORANGE; D2 · anode · GND · BLACK; CIN · positive · VIN · ORANGE; CIN · negative · GND · BLACK

Build — exact pin/net/color wiring

1. Connect F1 lead 1 to +8V_RAW using RED.
2. Connect F1 lead 2 to FUSED_8V using RED.
3. Connect D1 anode to FUSED_8V using RED.
4. Connect D1 cathode to VIN using ORANGE.
5. Connect D2 cathode to VIN using ORANGE.
6. Connect D2 anode to GND using BLACK.
7. Connect CIN positive to VIN using ORANGE.
8. Connect CIN negative to GND using BLACK.

Unpowered test

Instruction / kind / points / expected	Evidence
Check polarity and resistance · polarity · D1 cathode/VIN, D2 cathode/VIN, CIN negative/GND, VIN–GND · stripes correct and no short	

Powered test

Instruction / input / output / evidence / limits	Measured
Apply protected raw power · +8V_RAW at 8.00 V · FUSED_8V and VIN · record volts/current · VIN 7.20–7.90 V; coil-open current below 50 mA	

STOP gate: Current approaches 250 mA, VIN is outside 7.20–7.90 V, polarity is wrong, or anything warms. Remove power; do not build Cluster 2.

PASS gate: Protection chain, polarity, VIN window, and coil-open current pass. **PASS — continue** to Cluster 2 only.

Operator:	Date:	Signed PASS:

Cluster 2 — TSR supply

Dependencies: Cluster 1 **Inputs:** VIN, GND	**Outputs:** TSR_3V3 **Source state:** Cluster 1 source state retained; STANDALONE POWER remains removed

Parts

U2 TSR 1-2433E · 3.3 V converter · 1951-TSR1-2433E-ND · pin order verified. **C2/C3** 10 µF 50 V electrolytics · 1189-2322-ND · negative stripes to GND. **JP1** removable STANDALONE POWER · NOT ORDERED · leave removed.

Operator part record: U2/C2/C3/JP1 · TSR 1-2433E; two electrolytics; STANDALONE POWER · 3.3 V; 10 µF 50 V; removable link · 1951-TSR1-2433E-ND; 1189-2322-ND; NOT ORDERED · U2 pin order verified; capacitor negatives to GND; link removed
Operator wiring record: U2 · pin 1 · VIN · ORANGE; C2 · positive · VIN · ORANGE; U2 · pin 2 · GND · BLACK; C2 · negative · GND · BLACK; U2 · pin 3 · TSR_3V3 · BLUE; C3 · positive · TSR_3V3 · BLUE; C3 · negative · GND · BLACK; JP1 · between · TSR_3V3 and LOGIC_3V3 · BLUE

Build — exact pin/net/color wiring

1. Connect U2 pin 1 to VIN using ORANGE.
2. Connect C2 positive to VIN using ORANGE.
3. Connect U2 pin 2 to GND using BLACK.
4. Connect C2 negative to GND using BLACK.
5. Connect U2 pin 3 to TSR_3V3 using BLUE.
6. Connect C3 positive to TSR_3V3 using BLUE.
7. Connect C3 negative to GND using BLACK.
8. Fit JP1 between TSR_3V3 and LOGIC_3V3 using BLUE; leave JP1 removed.

Unpowered test

Instruction / kind / points / expected	Evidence
Verify converter map and capacitor polarity · continuity · U2 pins 1/2/3 to VIN/GND/TSR_3V3 · exact mapping; TSR_3V3 not joined to future LOGIC_3V3	

Powered test

Instruction / input / output / evidence / limits	Measured
Energize verified VIN · VIN · TSR_3V3 · record input/output/current · TSR_3V3 3.20–3.40 V and future LOGIC_3V3 unpowered	

STOP gate: TSR_3V3 is outside 3.20–3.40 V, polarity is wrong, LOGIC_3V3 is energized across the removed link, or current/temperature rises.

PASS gate: Verified Cluster 1 VIN produces isolated TSR_3V3 in range. **PASS — continue** to Cluster 3 only.

Operator:	Date:	Signed PASS:

Cluster 3 — DevKit and logic supply

Dependencies: Cluster 2; Verified USB bench source
Inputs: TSR_3V3, USB_5V, GND

Outputs: LOGIC_3V3, DevKit GND, GPIO21_CMD_POST, GPIO15_TX_ENABLE_POST, GPIO17_TX, GPIO38, USB_5V
Source state: First USB-only with STANDALONE POWER removed; then USB physically disconnected before standalone link installation

Parts

U1 ESP32-S3-DEVKITC-1-N8R8 · 1965-ESP32-S3-DEVKITC-1-N8R8-ND · antenna clear, USB ports identified. **JP1** STANDALONE POWER · NOT ORDERED · removable and labeled. **C4** 100 nF · BC1084CT-ND · non-polar.

Operator part record: U1/JP1/C4 · ESP32-S3-DEVKITC-1-N8R8; STANDALONE POWER; capacitor · DevKit; removable link; 100 nF · 1965-ESP32-S3-DEVKITC-1-N8R8-ND; NOT ORDERED; BC1084CT-ND · antenna clear; USB ports identified; link removable; capacitor non-polar
Operator wiring record: DevKit · 3V3 · LOGIC_3V3 · BLUE; DevKit · GND · GND · BLACK; C4 · lead 1 · LOGIC_3V3 · BLUE; C4 · lead 2 · GND · BLACK; TSR_3V3 · endpoint · the JP1 standalone-side post · BLUE; LOGIC_3V3 · endpoint · the JP1 logic-side post · BLUE; DevKit · GPIO15 · the DevKit-side GPIO15 jumper post · YELLOW; DevKit · GPIO21 · the DevKit-side GPIO21 jumper post · YELLOW

Build — exact pin/net/color wiring

1. Connect DevKit 3V3 to LOGIC_3V3 using BLUE.
2. Connect DevKit GND to GND using BLACK.
3. Connect C4 lead 1 to LOGIC_3V3 using BLUE.
4. Connect C4 lead 2 to GND using BLACK.
5. Connect TSR_3V3 endpoint to the JP1 standalone-side post using BLUE.
6. Connect LOGIC_3V3 endpoint to the JP1 logic-side post using BLUE; leave JP1 removed.
7. Connect DevKit GPIO15 to the DevKit-side GPIO15 jumper post using YELLOW; label the post.
8. Connect DevKit GPIO21 to the DevKit-side GPIO21 jumper post using YELLOW; label the post.

Unpowered test

Instruction / kind / points / expected	Evidence
Map source handoff · mapping · DevKit 3V3, LOGIC_3V3, TSR_3V3, JP1 · USB path mapped; JP1 removable; no source backfeed	

Powered test

Instruction / input / output / evidence / limits	Measured
Prove each exclusive source · USB_5V then TSR_3V3 · LOGIC_3V3 · record both runs · LOGIC_3V3 3.20–3.40 V; never energize USB and JP1 together	

STOP gate: Sources overlap, either source backfeeds the other, LOGIC_3V3 is outside 3.20–3.40 V, or DevKit boot is unstable.

PASS gate: USB-only and standalone-only handoff both pass, with the other source physically absent. **PASS — continue** to Cluster 4 only.

Operator: ____________ Date: ____________ Signed PASS: ____________

Cluster 4 — TPS3700 voltage monitor

Dependencies: Cluster 1, Cluster 3 **Inputs:** VIN, LOGIC_3V3, GND	**Outputs:** UV_SENSE, OV_SENSE, TREAD_OK, TREAD_OK_MCU **Source state:** USB logic only; STANDALONE POWER removed; bench source swept at protected VIN

Parts

U3 TPS3700DDCR · 296-30395-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **header** cut from PRPC040SAAN-RC · S1011EC-40-ND; **RUV** 150 kΩ · 150KXBK-ND; **ROV** 255 kΩ · 13-MFR-25FBF52-255K-ND; 10 kΩ/100 kΩ/4.7 kΩ · NOT ORDERED; two 1 nF · 445-173473-1-ND; 100 nF · BC1084CT-ND.

Adapter gate: continuity-map adapter pins 1–6 to the header before installation. Record the mapped header positions: 1 ____ 2 ____ 3 ____ 4 ____ 5 ____ 6 ____.

Divider formulas: UV_SENSE = VIN × 10 kΩ / (150 kΩ + 10 kΩ) = VIN / 16. OV_SENSE = VIN × 10 kΩ / (255 kΩ + 10 kΩ) = VIN / 26.5.

Operator part record: U3/dividers/filters · TPS3700DDCR plus adapter and UV/OV networks · 150 kΩ/10 kΩ; 255 kΩ/10 kΩ; 1 nF; 100 nF · 296-30395-1-ND; A879AR-ND; 150KXBK-ND; 13-MFR-25FBF52-255K-ND; 445-173473-1-ND; BC1084CT-ND; NOT ORDERED · meter-map adapter pin 1; capacitors non-polar
Operator wiring record: See numbered point-to-point steps below.

Operator header record: HDR3 · PRPC040SAAN-RC header · cut header · S1011EC-40-ND · map before installation

Build — exact pin/net/color wiring

1. Connect U3 pin 5 to VIN using ORANGE; record initials ____.
2. Connect U3 pin 2 to GND using BLACK; record initials ____.
3. Connect U3 pin 3 to UV_SENSE using WHITE; record initials ____.
4. Connect U3 pin 4 to OV_SENSE using WHITE; record initials ____.
5. Connect U3 pin 1 to TREAD_OK using YELLOW; record initials ____.
6. Connect U3 pin 6 to TREAD_OK using YELLOW; record initials ____.
7. Connect R_UV_TOP 1 to VIN using ORANGE; record initials ____.
8. Connect R_UV_TOP 2 to UV_SENSE using WHITE; record initials ____.
9. Connect R_UV_BOTTOM 1 to UV_SENSE using WHITE; record initials ____.
10. Connect R_UV_BOTTOM 2 to GND using BLACK; record initials ____.
11. Connect C_UV 1 to UV_SENSE using WHITE; record initials ____.
12. Connect C_UV 2 to GND using BLACK; record initials ____.
13. Connect R_OV_TOP 1 to VIN using ORANGE; record initials ____.
14. Connect R_OV_TOP 2 to OV_SENSE using WHITE; record initials ____.
15. Connect R_OV_BOTTOM 1 to OV_SENSE using WHITE; record initials ____.
16. Connect R_OV_BOTTOM 2 to GND using BLACK; record initials ____.
17. Connect C_OV 1 to OV_SENSE using WHITE; record initials ____.
18. Connect C_OV 2 to GND using BLACK; record initials ____.
19. Connect R_TREAD_PULLUP 1 to LOGIC_3V3 using BLUE; record initials ____.
20. Connect R_TREAD_PULLUP 2 to TREAD_OK using YELLOW; record initials ____.
21. Connect R_TREAD_PULLDOWN 1 to TREAD_OK using YELLOW; record initials ____.
22. Connect R_TREAD_PULLDOWN 2 to GND using BLACK; record initials ____.
23. Connect R_TREAD_MCU 1 to TREAD_OK using YELLOW; record initials ____.
24. Connect R_TREAD_MCU 2 to TREAD_OK_MCU using YELLOW; record initials ____.
25. Connect DEVKIT GPIO6 to TREAD_OK_MCU using YELLOW; record initials ____.
26. Connect C_TPS3700 1 to VIN using ORANGE; record initials ____.
27. Connect C_TPS3700 2 to GND using BLACK; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Meter divider and adapter · resistance · VIN–UV_SENSE–GND, VIN–OV_SENSE–GND, U3 pins · 150 kΩ/10 kΩ and 255 kΩ/10 kΩ networks; pin map exact	
Continuity-map adapter header · continuity · adapter pins 1–6 to header · one-to-one recorded mapping before installation	

Powered test

Instruction / input / output / evidence / limits	Measured
Sweep protected VIN · VIN · TREAD_OK · record low/high transitions · low below UV, high at 8.00 V, low above OV; UV rising 6.25–6.55 V; OV falling 10.30–10.90 V	

STOP gate: TREAD_OK three-state behavior or thresholds fail; never exceed protected VIN about 11.1 V; stop if TVS draws appreciable current or warms.

PASS gate: Both sweep directions and GPIO6 isolated observation match limits. **PASS — continue** to Cluster 5 only.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

Cluster 5 — AHC08 permission logic

Dependencies: Cluster 3, Cluster 4
Inputs: LOGIC_3V3, TREAD_OK, RELAY_CMD, TX_ENABLE, GND

Outputs: RELAY_GATE, TX_GATE
Source state: USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

Source state record: USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

Parts

U4 SN74AHC08N PDIP-14 · 296-4524-5-ND · notch verified, across center channel. **C5** 100 nF · BC1084CT-ND · non-polar. **R1/R2** 10 kΩ command pull-downs · NOT ORDERED.

Operator part record: U4/C5/R1/R2 · SN74AHC08N; capacitor; pull-downs · PDIP-14; 100 nF; two 10 kΩ · 296-4524-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar

Operator wiring record: See numbered point-to-point steps below.

Build — exact pin/net/color wiring

Complete removable endpoints: DevKit GPIO15 ↔ removable GPIO15 jumper ↔ TX_ENABLE / AHC08 pin 4. DevKit GPIO21 ↔ removable GPIO21 jumper ↔ RELAY_CMD / AHC08 pin 1.

1. Connect U4 pin 14 to LOGIC_3V3 using BLUE; record initials ____.
2. Connect U4 pin 7 to GND using BLACK; record initials ____.
3. Connect GPIO21 jumper post endpoint to U4 pin 1 / RELAY_CMD using YELLOW; record initials ____.
4. Connect U4 pin 2 to TREAD_OK using YELLOW; record initials ____.
5. Connect U4 pin 3 to RELAY_GATE using YELLOW; record initials ____.
6. Connect GPIO15 jumper post endpoint to U4 pin 4 / TX_ENABLE using YELLOW; record initials ____.
7. Connect U4 pin 5 to TREAD_OK using YELLOW; record initials ____.
8. Connect U4 pin 6 to TX_GATE using YELLOW; record initials ____.
9. Connect U4 9 to GND using BLACK; record initials ____.
10. Connect U4 10 to GND using BLACK; record initials ____.
11. Connect U4 12 to GND using BLACK; record initials ____.
12. Connect U4 13 to GND using BLACK; record initials ____.
13. Leave U4 pin 8 at NC using NO WIRE; record initials ____.
14. Leave U4 pin 11 at NC using NO WIRE; record initials ____.
15. Connect R_RELAY_CMD_PD 1 to RELAY_CMD using YELLOW; record initials ____.
16. Connect R_RELAY_CMD_PD 2 to GND using BLACK; record initials ____.
17. Connect R_TX_ENABLE_PD 1 to TX_ENABLE using YELLOW; record initials ____.
18. Connect R_TX_ENABLE_PD 2 to GND using BLACK; record initials ____.
19. Connect C_AHC08 1 to LOGIC_3V3 using BLUE; record initials ____.
20. Connect C_AHC08 2 to GND using BLACK; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Check DIP orientation and control mapping · mapping · U4 pins 1–14 and named nets · exact map, unused inputs grounded, outputs not shorted	

Powered test

Instruction / input / output / evidence / limits	Measured
Exercise both AND gates · RELAY_CMD/TX_ENABLE plus TREAD_OK · RELAY_GATE/TX_GATE · record 00,10,01,11 truth tables · output high only for command=1 and TREAD_OK=1	

Remove GPIO15 jumper before manual injection of TX_ENABLE. Remove GPIO21 jumper before manual injection of RELAY_CMD. Never drive against a connected DevKit GPIO.

STOP gate: Any gate rises without both inputs, a command GPIO remains connected during manual drive, or supply/reset behavior is abnormal.

PASS gate: Both four-state truth tables pass and temporary drives are removed. **PASS — continue** to Cluster 6 only.

Operator: ____________ Date: ____________ Signed PASS: ____________

Cluster 6 — TPS709 and BC337 driver

Dependencies: Cluster 1, Cluster 5
Inputs: VIN, RELAY_GATE, GND

Outputs: +5V_RLY, RELAY_COIL−
Source state: Valid VIN/TREAD_OK; COIL POWER remains removed for the unloaded test

Parts

U5 TPS70950DBVR · 296-35484-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **header** cut from PRPC040SAAN-RC · S1011EC-40-ND; **Q1** BC337-40 · 4878-BC337-40CT-ND; **RBASE** 560 Ω · MFR-25FBF52-560R-ND; 10 kΩ · NOT ORDERED; 1 µF · 445-173583-1-ND; 4.7 µF · 732-8660-1-ND; **JP2** COIL POWER · NOT ORDERED.

Operator part record: U5/Q1/RBASE/caps/JP2 · TPS70950DBVR; BC337-40; 560 Ω/10 kΩ; 1 µF/4.7 µF; COIL POWER · 5 V regulator and unloaded driver · 296-35484-1-ND; A879AR-ND; 4878-BC337-40CT-ND; MFR-25FBF52-560R-ND; 445-173583-1-ND; 732-8660-1-ND; NOT ORDERED · adapter and Q1 leads mapped; electrolytic negative to GND; link removed

Operator wiring record: See numbered point-to-point steps below.

Operator header record: HDR5 · PRPC040SAAN-RC header · cut header · S1011EC-40-ND · map before installation

Build — exact pin/net/color wiring

1. Connect U5 pin 1 to VIN using ORANGE; record initials ____.
2. Connect U5 pin 2 to GND using BLACK; record initials ____.
3. Connect U5 pin 3 to RELAY_GATE using YELLOW; record initials ____.
4. Connect U5 pin 5 to +5V_RLY using VIOLET; record initials ____.
5. Leave U5 pin 4 at NC using NO WIRE; record initials ____.
6. Connect C_TPS709_IN 1 to VIN using ORANGE; record initials ____.
7. Connect C_TPS709_IN 2 to GND using BLACK; record initials ____.
8. Connect C_TPS709_OUT positive to +5V_RLY using VIOLET; record initials ____.
9. Connect C_TPS709_OUT negative to GND using BLACK; record initials ____.
10. Connect R_BASE 1 to RELAY_GATE using YELLOW; record initials ____.
11. Connect R_BASE 2 to Q_BASE using YELLOW; record initials ____.
12. Connect Q1 base to Q_BASE using YELLOW; record initials ____.
13. Connect R_BASE_PD 1 to Q_BASE using YELLOW; record initials ____.
14. Connect R_BASE_PD 2 to GND using BLACK; record initials ____.
15. Connect Q1 emitter to GND using BLACK; record initials ____.
16. Connect Q1 collector to RELAY_COIL− using VIOLET; record initials ____.
17. Connect JP2 supply side to +5V_RLY using VIOLET; record initials ____.
18. Connect JP2 coil side to RELAY_COIL+ using VIOLET; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Verify adapter, transistor leads, and isolation · polarity · U5 pins, Q1 B/C/E, output capacitor, JP2 · exact map; COIL POWER open	

Powered test

Instruction / input / output / evidence / limits	Measured
Test gate unloaded · RELAY_GATE · +5V_RLY/Q1 base · record on/off · TPS709 enabled 4.75–5.25 V; TPS709 disabled below 0.25 V	

STOP gate: COIL POWER is installed, enabled or disabled limits fail, output does not discharge below 0.25 V, or Q1/U5 heats.

PASS gate: Unloaded regulator and base drive pass with coil isolated. **PASS — continue** to Cluster 7 only.

Operator: ______ Date: ______ Signed PASS: ______

Cluster 7 — Relay coil, local contacts, and feedback

Dependencies: Cluster 6, Cluster 3
Inputs: +5V_RLY, RELAY_COIL−, LOGIC_3V3, GND

Outputs: LOCAL_NC, LOCAL_NO, FB_NC, FB_NO
Source state: Local-only bench exercise; treadmill/RJ45 cables absent; bounded relay exerciser verified; loaded relay limit no more than 500 mA

Parts

K1 G5V-2 DC5 DPDT · 39-G5V-2DC5-ND · bottom-view drawing, meter-map physical pins. **D3** P6KE6.8CA bidirectional · 18-P6KE6.8CACT-ND. **JP2** COIL POWER · NOT ORDERED. **RFB** two 10 kΩ · NOT ORDERED.

Operator part record: K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 kΩ · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional
Operator wiring record: See numbered point-to-point steps below.

Build — exact pin/net/color wiring

1. Connect K1 pin 1 to RELAY_COIL+ using VIOLET; record initials ____.
2. Connect K1 pin 16 to RELAY_COIL− using VIOLET; record initials ____.
3. Connect D3 lead 1 to RELAY_COIL+ using VIOLET; record initials ____.
4. Connect D3 lead 2 to RELAY_COIL− using VIOLET; record initials ____.
5. Connect K1 pin 4 to LOCAL_NC using WHITE; record initials ____.
6. Connect K1 pin 6 to LOCAL_TX_SELECTED using WHITE; record initials ____.
7. Connect K1 pin 8 to LOCAL_NO using WHITE; record initials ____.
8. Connect K1 pin 11 to GND using BLACK; record initials ____.
9. Connect K1 pin 13 to K1_NC_FB using GREEN; record initials ____.
10. Connect DEVKIT GPIO4 to K1_NC_FB using GREEN; record initials ____.
11. Connect K1 pin 9 to K1_NO_FB using GREEN; record initials ____.
12. Connect DEVKIT GPIO5 to K1_NO_FB using GREEN; record initials ____.
13. Connect R_FB_NC 1 to LOGIC_3V3 using BLUE; record initials ____.
14. Connect R_FB_NC 2 to K1_NC_FB using GREEN; record initials ____.
15. Connect R_FB_NO 1 to LOGIC_3V3 using BLUE; record initials ____.
16. Connect R_FB_NO 2 to K1_NO_FB using GREEN; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Map relay before attaching local fixture · continuity · pins 1–16, 4–6, 8–6, 13–11, 9–11 · coil about 50 Ω; NC pairs closed; NO pairs open	

Powered test

Instruction / input / output / evidence / limits	Measured
Exercise bounded relay and fail sources · +5V_RLY/RELAY_COIL− · local contacts/feedback · record current, voltage, VCE, timing, temperature · coil 90–110 mA, voltage at least 4.50 V, BC337 VCE no more than 0.30 V; (1,0) energized, (0,1) bypass; 00 or 11 fault	

Bound operator part record: K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 kΩ · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional

Relay exerciser identity: ____________________ **Firmware identity:** ____________________

From energized state, removal of USB logic power must release to NC bypass; removal of GPIO21 command must release to NC bypass; removal of TREAD_OK must release to NC bypass; removal of VIN must release to NC bypass. Each NC bypass return must occur in no more than 100 ms. Then hold five minutes: TPS709 and BC337 at or below 45°C and no more than 10°C over ambient.

Dedicated fail-release and thermal evidence

Evidence field	Measured	Limit
USB logic release (ms)		NC bypass ≤100 ms
GPIO21 release (ms)		NC bypass ≤100 ms
TREAD_OK release (ms)		NC bypass ≤100 ms
VIN release (ms)		NC bypass ≤100 ms
Ambient temperature (°C)		Record before hold
TPS709 temperature (°C)		≤45°C and ≤10°C over ambient
BC337 temperature (°C)		≤45°C and ≤10°C over ambient

STOP gate: Any electrical, feedback, release, reset, or five-minute thermal gate fails. Cluster 7 is local-only; it makes no end-to-end RJ45 claim.

PASS gate: Local contacts, feedback, current, voltage, release, and thermal evidence all pass. **PASS — continue** to Cluster 8 only.

Operator: ____________ Date: ____________ Signed PASS: ____________

Cluster 8 — AHC126 and UART taps

Dependencies: Cluster 3, Cluster 5, Cluster 7
Inputs: LOGIC_3V3, TX_GATE, GPIO17_TX, LOCAL_NC, LOCAL_NO, GND

Outputs: TX_DRV, UART_BUFFER_HIGH_Z, UART_BUFFER_FOLLOWS, GPIO18_RX_LOCAL, GPIO16_RX_LOCAL, LOCAL_TX_SELECTED
Source state: Local future-interface fixture only; treadmill and both RJ45 breakouts absent; bounded UART/relay exerciser identity recorded

Parts

U6 SN74AHC126N PDIP-14 · 296-4535-5-ND · notch verified. **C6** 100 nF · BC1084CT-ND. **R3** 100 Ω and **R4/R5** 10 kΩ · NOT ORDERED.

Operator part record: U6/C6/R3/R4/R5 · SN74AHC126N; capacitor; series/tap resistors · PDIP-14; 100 nF; 100 Ω; two 10 kΩ · 296-4535-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar
Operator wiring record: See numbered point-to-point steps below.

Build — exact pin/net/color wiring

Local future-interface fixture only: use a bounded UART/relay exerciser; no RJ45 or treadmill cable.

1. Connect U6 pin 14 to LOGIC_3V3 using BLUE; record initials ____.
2. Connect U6 pin 7 to GND using BLACK; record initials ____.
3. Connect U6 pin 1 to TX_GATE using YELLOW; record initials ____.
4. Connect U6 pin 2 to GPIO17_TX using YELLOW; record initials ____.
5. Connect U6 pin 3 to TX_BUF using YELLOW; record initials ____.
6. Connect R_TX 1 to TX_BUF using YELLOW; record initials ____.
7. Connect R_TX 2 to TX_DRV using YELLOW; record initials ____.
8. Connect U6 4 to GND using BLACK; record initials ____.
9. Connect U6 5 to GND using BLACK; record initials ____.
10. Connect U6 9 to GND using BLACK; record initials ____.
11. Connect U6 10 to GND using BLACK; record initials ____.
12. Connect U6 12 to GND using BLACK; record initials ____.
13. Connect U6 13 to GND using BLACK; record initials ____.
14. Leave U6 pin 6 at NC using NO WIRE; record initials ____.
15. Leave U6 pin 8 at NC using NO WIRE; record initials ____.
16. Leave U6 pin 11 at NC using NO WIRE; record initials ____.
17. Connect C_AHC126 1 to LOGIC_3V3 using BLUE; record initials ____.
18. Connect C_AHC126 2 to GND using BLACK; record initials ____.
19. Connect R_CONS_RX 1 to LOCAL_CONSOLE_6_STUB using WHITE; record initials ____.
20. Connect R_CONS_RX 2 to GPIO18_RX_LOCAL using GREEN; record initials ____.
21. Connect DEVKIT GPIO18 to GPIO18_RX_LOCAL using GREEN; record initials ____.
22. Connect R_PIN3_RX 1 to LOCAL_PIN3_STUB using WHITE; record initials ____.
23. Connect R_PIN3_RX 2 to GPIO16_RX_LOCAL using GREEN; record initials ____.
24. Connect DEVKIT GPIO16 to GPIO16_RX_LOCAL using GREEN; record initials ____.
25. Connect K1 pin 4 / LOCAL_NC to LOCAL_CONSOLE_6_STUB using WHITE; record initials ____.
26. Connect K1 pin 8 / LOCAL_NO to TX_DRV using YELLOW; record initials ____.
27. Connect K1 pin 6 to LOCAL_TX_SELECTED using YELLOW; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Check local buffer map · continuity · U6 pins, 100 Ω resistor, local TX_DRV test point · exact map; disabled output isolated; no connector-side wiring	
Verify GPIO18 local receive tap · resistance · LOCAL_CONSOLE_6_STUB → GPIO18_RX_LOCAL · 10 kΩ nominal (9–11 kΩ); isolated from LOGIC_3V3, GND, and TX_DRV	
Verify GPIO16 local receive tap · resistance · LOCAL_PIN3_STUB → GPIO16_RX_LOCAL · 10 kΩ nominal (9–11 kΩ); isolated from LOGIC_3V3, GND, and TX_DRV	

Powered test

Instruction / input / output / evidence / limits	Measured
Exercise local buffer enable · TX_GATE/GPIO17_TX/local fixture · TX_DRV/UART_BUFFER_HIGH_Z/UART_BUFFER_FOLLOWS · capture logic traces · pin 3 high impedance disabled and follows pin 2 enabled	
Stimulate GPIO18 local receive tap · LOCAL_CONSOLE_6_STUB low/high from bounded UART exerciser · GPIO18_RX_LOCAL · record low/high voltage and observed bits · low ≤0.40 V; high ≥2.40 V; bits follow	
Stimulate GPIO16 local receive tap · LOCAL_PIN3_STUB low/high from bounded UART exerciser · GPIO16_RX_LOCAL · record low/high voltage and observed bits · low ≤0.40 V; high ≥2.40 V; bits follow	
Exercise bounded local relay transfer · TX_DRV/K1 LOCAL_NC/LOCAL_NO with bounded relay exerciser · LOCAL_TX_SELECTED · record NC/NO selected resistance and TX waveform · selected path below 2 Ω; unselected path open; waveform follows TX_DRV	

Relay exerciser identity: ____________________ **Firmware identity:** ____________________

Dedicated local future-interface evidence

Evidence field	Measured	Limit/state
GPIO18_RX_LOCAL low (V)		≤0.40 V
GPIO18_RX_LOCAL high (V)		≥2.40 V
GPIO16_RX_LOCAL low (V)		≤0.40 V
GPIO16_RX_LOCAL high (V)		≥2.40 V
LOCAL_TX_SELECTED NC (Ω)		<2 Ω selected
LOCAL_TX_SELECTED NO (Ω)		<2 Ω selected

STOP gate: TX_DRV drives while disabled, enabled waveform does not follow, either receive tap fails, local relay selection fails, any connector-side wire is installed, or exerciser identity is absent.

PASS gate: Local high impedance, following, receive taps, and relay-selected TX pass; connector-side checks remain deferred. **PASS — continue** to Cluster 9 only.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

Cluster 9 — Indicators and VBUS sensing

Dependencies: Cluster 3 **Inputs:** LOGIC_3V3, GPIO38, USB_5V, GND	**Outputs:** POWER_LED, STATUS_LED, VBUS_PRESENT_N **Source state:** USB-only test then standalone-only test; never join VBUS to a local rail

Parts

Q2 2N7000 · 4878-2N7000CT-ND; **D4** green LTL-4233 · 160-1130-ND; **D5** red 151051RS11000 · 732-5016-ND; 330 Ω, 1 kΩ, and 10 kΩ resistors · NOT ORDERED; 100 nF · BC1084CT-ND.

Operator part record: Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330 Ω/1 kΩ/10 kΩ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode
Operator wiring record: See numbered point-to-point steps below.

Operator bypass wiring record: C7 · leads 1/2 · LOGIC_3V3/GND · BLUE/BLACK

Build — exact pin/net/color wiring

Bypass summary: C7 lead 1 to LOGIC_3V3 and C7 lead 2 to GND.

1. Connect DEVKIT 5V/VBUS to VBUS_SENSE using VIOLET; record initials ____.
2. Connect Q2 gate to VBUS_SENSE using VIOLET; record initials ____.
3. Connect R_VBUS_DISCHARGE 1 to VBUS_SENSE using VIOLET; record initials ____.
4. Connect R_VBUS_DISCHARGE 2 to GND using BLACK; record initials ____.
5. Connect Q2 source to GND using BLACK; record initials ____.
6. Connect Q2 drain to VBUS_PRESENT_N using GREEN; record initials ____.
7. Connect DEVKIT GPIO7 to VBUS_PRESENT_N using GREEN; record initials ____.
8. Connect R_VBUS_PULLUP 1 to LOGIC_3V3 using BLUE; record initials ____.
9. Connect R_VBUS_PULLUP 2 to VBUS_PRESENT_N using GREEN; record initials ____.
10. Connect DEVKIT GPIO38 to STATUS_LED using YELLOW; record initials ____.
11. Connect R_STATUS 1 to STATUS_LED using YELLOW; record initials ____.
12. Connect R_STATUS 2 to LED_GREEN_A using YELLOW; record initials ____.
13. Connect LED_GREEN anode to LED_GREEN_A using YELLOW; record initials ____.
14. Connect LED_GREEN cathode to GND using BLACK; record initials ____.
15. Connect R_POWER_LED 1 to LOGIC_3V3 using BLUE; record initials ____.
16. Connect R_POWER_LED 2 to LED_RED_A using BLUE; record initials ____.
17. Connect LED_RED anode to LED_RED_A using BLUE; record initials ____.
18. Connect LED_RED cathode to GND using BLACK; record initials ____.
19. Connect C_LOGIC 1 to LOGIC_3V3 using BLUE; record initials ____.
20. Connect C_LOGIC 2 to GND using BLACK; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Verify transistor leads and LED polarity · polarity · Q2 G/D/S, both LED anodes/cathodes · exact map; VBUS isolated from all local positive rails	

Powered test

Instruction / input / output / evidence / limits	Measured
Toggle source and status · USB_5V/LOGIC_3V3/GPIO38 · VBUS_PRESENT_N/LEDs · record truth/voltage · VBUS sense active-low; USB present low, USB absent high; red power and green status indications correct	

Bound operator part record: Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330 Ω/1 kΩ/10 kΩ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode

VBUS sense is active-low and isolated: no connection may ever join VBUS to a local rail.

STOP gate: VBUS joins a local rail, active-low truth is wrong, LED polarity/current limiting is wrong, or sources overlap.

PASS gate: Both LEDs and isolated active-low VBUS sense pass under exclusive sources. **PASS — continue** to Cluster 10 only.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

Cluster 10 — Whole-device standalone bench test

Dependencies: Cluster 1, Cluster 2, Cluster 3, Cluster 4, Cluster 5, Cluster 6, Cluster 7, Cluster 8, Cluster 9 **Inputs:** +8V_RAW, GND, observer firmware	**Outputs:** standalone boot, Wi-Fi observation, event log, UART idle evidence, verified observer state **Source state:** USB physically absent; STANDALONE POWER installed; observer verified; relay off; TX disabled

Outputs record: standalone boot · Wi-Fi observation · event log · UART idle evidence

Parts

ASSEMBLY completed Clusters 1–9 · NOT ORDERED · inspect all removable links and polarities. **Firmware artifact** observer image and manifest · NOT ORDERED · exact immutable identity required.

Operator part record: ASSEMBLY/FW · completed Clusters 1–9; observer image and manifest · whole device; immutable diagnostic artifact · NOT ORDERED · inspect links/polarities; observer manifest proves relay and TX disabled
Operator wiring record: USB · plug · DISCONNECTED · NO WIRE; STANDALONE POWER · between · TSR_3V3 and LOGIC_3V3 · BLUE; bench · positive lead · +8V_RAW · RED; bench · return lead · GND · BLACK

Build — exact pin/net/color wiring

1. Connect USB plug to DISCONNECTED using NO WIRE.
2. Install STANDALONE POWER between TSR_3V3 and LOGIC_3V3 using BLUE.
3. Connect the bench positive lead to +8V_RAW using RED.
4. Connect the bench return lead to GND using BLACK.

Unpowered test

Instruction / kind / points / expected	Evidence
Final standalone source audit · resistance · +8V_RAW/VIN/TSR_3V3/LOGIC_3V3/+5V_RLY to GND · no shorts; USB absent; source handoff exact	

Powered test

Instruction / input / output / evidence / limits	Measured
Boot observer from raw input · +8V_RAW · standalone boot/Wi-Fi/event log/UART idle · record rails, reset log, GPIO16/GPIO18 idle · VIN 7.20–7.90 V; TSR_3V3 and LOGIC_3V3 3.20–3.40 V; relay off, feedback (0,1), TX disabled	

Firmware identity: ____________________ **Observer identity:** ____________________

Observer manifest: ____________________ (path or SHA proving relay and TX outputs disabled)

Dedicated UART idle evidence

Evidence field	Measured	State
GPIO16 UART idle (V)		Before UART configuration
GPIO18 UART idle (V)		Before UART configuration

STOP gate: Identity/manifest missing, rail out of range, reset, Wi-Fi/event-log failure, UART idle missing, relay enables, TX enables, or feedback differs from (0,1).

PASS gate: Identified observer boots standalone and all passive observations pass. **PASS — continue** to Cluster 11 only.

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

Cluster 11 — RJ45 pass-through and treadmill bypass

Cluster 11 is the end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass proof.

Dependencies: Cluster 7, Cluster 8, Cluster 10 **Inputs:** CONSOLE.1–8, MOTOR.1–8, verified observer state	**Outputs:** eight mapped conductors, end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass, bypass-only treadmill evidence **Source state:** USB absent; standalone installed; observer verified; relay off; TX disabled; source → state → harness → direct-path sequence

Parts

J1/J2 CONSOLE/MOTOR RJ45 breakouts · NOT ORDERED · meter-map terminal numbers. **Harness** reviewed dual-pin fused-DMM current harness · NOT ORDERED. **K1** verified Cluster 7 relay contacts · 39-G5V-2DC5-ND.

Operator part record: J1/J2/HARNESS/K1 · CONSOLE/MOTOR RJ45 breakouts; dual-pin fused-DMM harness; verified relay · two 8P8C breakouts; reviewed current harness; G5V-2 · NOT ORDERED; 39-G5V-2DC5-ND · meter-map connector terminals; verify fused jack/polarity; relay bottom-view map proven

Operator wiring record: J1/J2 · terminals · unwired · NO WIRE; J1 · pin 1 · GND · BLACK; J2 · pin 1 · GND · BLACK; J1 · pin 7 · GND · BLACK; J2 · pin 7 · GND · BLACK; J1 · pin 2 · +8V_RAW · RED; J2 · pin 2 · +8V_RAW · RED; J1 · pin 8 · +8V_RAW · RED; J2 · pin 8 · +8V_RAW · RED; CONSOLE.3 · endpoint · MOTOR.3 · WHITE; CONSOLE.4 · endpoint · MOTOR.4 · WHITE; CONSOLE.5 · endpoint · MOTOR.5 · WHITE; CONSOLE.6 · endpoint · K1 pin 4 · YELLOW; MOTOR.6 · endpoint · K1 pin 6 · YELLOW; K1 · pin 8 · TX_DRV · YELLOW; CONSOLE.6 · through 10 kΩ · GPIO18_RX · GREEN; PIN3 · through 10 kΩ · GPIO16_RX · GREEN

Build — exact pin/net/color wiring

1. Isolated mapping first: keep all J1/J2 terminals unwired; map each conductor independently and verify unrelated pins open using NO WIRE.

Independent isolated-map evidence — complete before Step 2

CONSOLE terminal map	MOTOR terminal map	Unrelated pins open / operator initials
1 ___ 2 ___ 3 ___ 4 ___ 5 ___ 6 ___ 7 ___ 8 ___	1 ___ 2 ___ 3 ___ 4 ___ 5 ___ 6 ___ 7 ___ 8 ___	

Common only after isolated-map PASS.

2. Connect J1 pin 1 to GND using BLACK; this begins the join pins 1/7 and 2/8 commoning sequence.
3. Connect J2 pin 1 to GND using BLACK.
4. Connect J1 pin 7 to GND using BLACK.
5. Connect J2 pin 7 to GND using BLACK.
6. Connect J1 pin 2 to +8V_RAW using RED.
7. Connect J2 pin 2 to +8V_RAW using RED.
8. Connect J1 pin 8 to +8V_RAW using RED.
9. Connect J2 pin 8 to +8V_RAW using RED.
10. Connect CONSOLE.3 endpoint to MOTOR.3 using WHITE.
11. Connect CONSOLE.4 endpoint to MOTOR.4 using WHITE.
12. Connect CONSOLE.5 endpoint to MOTOR.5 using WHITE.
13. Connect CONSOLE.6 endpoint to K1 pin 4 using YELLOW.
14. Connect MOTOR.6 endpoint to K1 pin 6 using YELLOW.
15. Connect K1 pin 8 to TX_DRV using YELLOW.
16. Connect CONSOLE.6 through 10 kΩ to GPIO18_RX using GREEN.
17. Connect PIN3 through 10 kΩ to GPIO16_RX using GREEN.

Unpowered test

Instruction / kind / points / expected	Evidence
Map every connector conductor independently · mapping · CONSOLE.1–8 to MOTOR.1–8 and K1 pins 4/6/8 · each intended path below 2 Ω; post-common cross-shorts absent except pins 1/7 GND and pins 2/8 +8V_RAW; end-to-end CONSOLE.6 ↔ MOTOR.6 through NC	

Powered test

Instruction / input / output / evidence / limits	Measured
Run bypass-only treadmill sequence · treadmill +8 V through fused harness · current/drop/receive-only UART/thermal · record source state, harness, restored direct paths, endpoints · separately measure treadmill current no more than 500 mA; supply drop no more than 50 mV; ground-return drop no more than 50 mV; 15-minute endpoints no more than 40°C and no more than 10°C over ambient	

Firmware identity: ____________________ **Observer identity:** ____________________

Observer manifest: ____________________ (path or SHA proving relay and TX outputs disabled)

Bypass-only controlled sequence

1. All sources off before RJ45 changes. Attach cables only after Clusters 1–10 signed PASS.
2. Verify bypass-only source state: USB absent; STANDALONE POWER installed; observer verified; relay off; TX disabled.
3. With all sources off before harness changes, remove both independent direct +8V paths and install the reviewed dual-pin fused-DMM harness. Measure current separately at no more than 500 mA.
4. Remove power before removal of the harness. Verify zero current, remove harness, then restore both independent +8V paths and continuity-check each.
5. Voltage drop is measured only after direct-path restoration. Thermal test only after direct-path restoration; then run the fifteen-minute hold.

Cluster 11 — dedicated bypass and thermal evidence

Evidence field	Measured	Limit/state
Treadmill current (mA)		≤500 mA, fused harness
Supply drop (mV)		≤50 mV after direct paths restored
Ground-return drop (mV)		≤50 mV after direct paths restored

RJ45 pin	CONSOLE temperature (°C)	MOTOR temperature (°C)	Limit after 15 minutes
1 — GND			≤40°C; ≤10°C rise
2 — +8 V			≤40°C; ≤10°C rise
3			≤40°C; ≤10°C rise
4			≤40°C; ≤10°C rise
5			≤40°C; ≤10°C rise
6 — switched data			≤40°C; ≤10°C rise
7 — GND			≤40°C; ≤10°C rise
8 — +8 V			≤40°C; ≤10°C rise

Relevant breadboard power endpoint	Measured	Limit
+8V endpoint temperature (°C): hottest exact endpoint ___________		≤40°C; ≤10°C rise
GND endpoint temperature (°C): hottest exact endpoint ___________		≤40°C; ≤10°C rise

The current no-control diagnostic cannot pass the relay exercise gate and cannot pass the UART exercise gate. This is observation only. No treadmill transfer and no MCU TX. A future procedure must require qualified functional-firmware identity and matching production safety evidence; bypass PASS grants no transfer permission.

STOP gate: Any map, identity, current, drop, receive-only UART, bypass, thermal, or reset check fails. Power off; do not transfer the relay or transmit.

PASS gate: All eight paths, NC bypass, current, restored-path drop, receive-only observation, and thermal evidence pass. **PASS — continue** only to documentation closeout, never to treadmill control.

Operator: ______________________ Date: ______________________ Signed PASS: ______________________